

Adipogenic Human Adenovirus Ad-36 Induces Commitment, Differentiation, and Lipid Accumulation in Human Adipose-Derived Stem Cells

Human adenovirus Ad-36 is causatively and correlatively linked with animal and human obesity, respectively. Ad-36 enhances differentiation of rodent preadipocytes, but its effect on adipogenesis in humans is unknown. To indirectly assess the role of Ad-36-induced adipogenesis in human obesity, the effect of the virus on commitment, differentiation, and lipid accumulation was investigated in vitro in primary human adipose-derived stem/stromal cells (hASC). Ad-36 infected hASC in a time- and dose-dependent manner. Even in the presence of osteogenic media, Ad-36-infected hASC showed significantly greater lipid accumulation, suggestive of their commitment to the adipocyte lineage. Even in the absence of adipogenic inducers, Ad-36 significantly increased hASC differentiation, as indicated by a time-dependent expression of genes within the adipogenic cascade—CCAAT/Enhancer binding protein- β , peroxisome proliferator-activated receptor- γ , and fatty acid-binding protein—and consequentially increased lipid accumulation in a time- and viral dose-dependent manner. Induction of hASC to the adipocyte state by Ad-36 was further supported by increased expression of lipoprotein lipase and the accumulation of its extracellular fraction. hASC from subjects harboring Ad-36 DNA in their adipose tissue due to natural infection had significantly greater ability to differentiate compared with Ad-36 DNA-negative counterparts, which offers a proof of concept. Thus, Ad-36 has the potential to induce adipogenesis in hASC, which may contribute to adiposity induced by the virus.